Guide 57 — Medical Equipment Repairer

Controlled English working master — Version 2.0
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What this work is

Medical equipment repairers install, inspect, maintain, troubleshoot, calibrate, and repair equipment used in patient care. Employers may use titles such as **medical equipment repairer**, **biomedical equipment technician (BMET)**, **biomedical technician**, **clinical engineering technician**, **field service technician**, or similar titles. Titles and scope vary by employer and jurisdiction.

This occupation sits at the intersection of electronics, mechanics, software, networking, quality systems, safety, and healthcare operations. The work can include simple devices as well as complex systems that affect diagnosis, monitoring, therapy, or life support.

A job title never gives permission to perform every possible task. Work only within your training, employer authorization, manufacturer procedures, applicable regulation, and the specific service scope assigned to you.

What a medical equipment repairer may do

Depending on training, authorization, equipment class, and employer, typical work may include:

- receiving and inspecting new equipment;
- installing equipment and verifying basic operation;
- performing scheduled preventive maintenance;
- inspecting cords, connectors, batteries, sensors, housings, controls, alarms, accessories, and other components;
- using approved test equipment to verify electrical, mechanical, environmental, or performance parameters;
- troubleshooting faults using service documentation and approved diagnostic tools;
- replacing authorized parts and assemblies;
- calibrating or verifying equipment according to approved procedures;
- documenting maintenance, parts, measurements, software/firmware state, corrective actions, and return-to-service results;
- supporting equipment inventory, recall, alert, and service-history records;

- coordinating with manufacturers, vendors, clinical engineering, information technology, cybersecurity, facilities, infection prevention, and clinical users;
- helping identify repeated failures, safety hazards, or abnormal trends; and
- escalating incidents, suspected defects, cybersecurity concerns, or work outside authorized service procedures.

Some roles involve field travel, hospital rounds, shop work, bench repair, networked-device support, or on-call coverage.

What this role must not assume

Do not independently:

- diagnose a patient or make treatment decisions;
- change clinical settings because you believe a different therapy would be better;
- disable safety alarms, interlocks, grounding protections, access controls, or other safeguards merely to make equipment operate;
- install unapproved parts, firmware, software, patches, drivers, or configuration changes;
- alter a device's intended use or materially change safety/performance specifications without the required engineering, quality, manufacturer, and regulatory process;
- bypass required calibration, test, documentation, infection-control, or return-to-service steps;
- falsify service records, test results, serial numbers, parts, incident information, or maintenance dates;
- use another worker's credentials or share service passwords; or
- hide a malfunction or adverse event that may have affected patient or operator safety.

When a task may cross from ordinary servicing into a modification or remanufacturing activity, stop and escalate before proceeding.

Servicing versus remanufacturing — United States

The U.S. Food and Drug Administration distinguishes **servicing** from **remanufacturing**.

In FDA's framework, servicing generally means repair, preventive maintenance, or routine maintenance intended to return a finished device to the original equipment manufacturer's safety and performance specifications and original intended use.

Remanufacturing can include activity that significantly changes a finished device's performance or safety specifications or intended use. The label a company gives the work does not control the regulatory classification; what the activity actually does to the device matters.

For an individual technician, the practical rule is simple: **do not improvise a design change under the label of repair**. If a proposed change is outside approved service instructions or could materially affect safety, performance, cybersecurity, labeling, or intended use, escalate to the responsible engineering, quality, manufacturer, and regulatory function.

Patient and worker safety

Medical-equipment service can expose workers to serious hazards. Depending on the device and environment, risks may include:

- line voltage and stored electrical energy;
- high-voltage capacitors;
- batteries and thermal runaway;
- moving mechanical parts;
- compressed gas or pressure systems;
- hot surfaces;
- lasers or optical radiation;
- ionizing-radiation environments;
- magnetic-field hazards;
- biological contamination;
- sharps or body-fluid exposure;
- hazardous chemicals;
- heavy or awkward equipment;
- trip and ergonomic hazards; and
- unexpected device activation.

Use the employer's lockout/tagout or energy-control procedure where applicable. Follow required personal protective equipment, infection-prevention, decontamination, electrical-safety, radiation-safety, lifting, and equipment-isolation practices.

Do not service contaminated equipment as though it were an ordinary consumer electronic product. Confirm the required decontamination and infection-control process before opening or transporting equipment when exposure is possible.

Return-to-service discipline

Repair is not complete merely because a device powers on.

Before return to service, follow the applicable procedure for:

- functional verification;
- calibration or performance verification where required;
- alarm and safety checks;
- electrical-safety testing where applicable;
- accessory and battery verification;
- software/firmware/configuration confirmation;
- labels, seals, covers, fasteners, and guards;
- cleaning/decontamination status;
- documentation of parts and measurements;
- unresolved limitations or exceptions; and
- required review or sign-off.

If a device cannot meet the required acceptance criteria, keep it out of service and escalate rather than changing the acceptance criteria to make the repair pass.

Documentation and traceability

High-quality service records help protect patients, workers, employers, manufacturers, and future technicians.

Document information required by the employer and procedure, which may include:

- asset number and serial number;
- model and manufacturer;
- service request or work-order number;
- reported problem;
- inspection findings;
- parts replaced;
- measurements and test equipment used;
- calibration or verification results;
- firmware/software/configuration state when relevant;
- corrective action;
- unresolved concerns;
- incident or alert linkage; and
- return-to-service status.

Never backdate, invent, copy-forward, or alter results to make a record look complete.

Medical-device cybersecurity

Modern devices may be networked computers as well as clinical equipment. A repairer may encounter network addresses, wireless configuration, service accounts, firmware, logs, removable media, vendor remote-access systems, certificates, cryptographic material, or cloud-connected features.

Safe servicing principles include:

- use only approved service accounts, tools, laptops, media, and remote-access methods;
- never share passwords or service credentials;
- do not connect unknown USB devices or personal equipment to medical systems;
- verify firmware/software packages and approved versions before installation;
- follow change-management requirements for network or configuration changes;
- preserve logs and evidence if a cybersecurity incident is suspected;
- escalate unexpected network behavior, unauthorized access, malware indicators, unusual reboots, unexplained configuration changes, or security alerts;
- follow manufacturer and employer vulnerability-response procedures; and
- coordinate with cybersecurity/IT rather than treating a suspected compromise as an ordinary hardware fault.

Cybersecurity responsibility does not mean a repairer should perform unauthorized penetration testing on clinical devices.

Privacy and confidential information

A technician may encounter patient names, identifiers, images, waveforms, reports, schedules, or other protected health information while testing or servicing equipment.

Use the minimum information necessary for authorized work. Follow the employer's privacy and security rules. Do not photograph screens, copy patient data, export logs, or move data to personal storage unless explicitly authorized and required by procedure.

Never place patient information, credentials, proprietary service manuals, device secrets, internal network data, or confidential configuration material into a public or unapproved generative-AI system.

Responsible use of AI

AI can sometimes help with general learning, terminology, study planning, or drafting non-sensitive checklists. It must not replace:

- manufacturer service instructions;
- approved test procedures;

- engineering judgment for safety-critical changes;
- employer policy;
- regulatory requirements;
- cybersecurity incident response;
- calibration standards;
- clinical judgment; or
- emergency escalation.

Treat AI output as unverified until checked against approved sources. Never let an AI system persuade you to bypass a safety interlock, change a clinical setting, invent a test result, or use an unapproved part or firmware image.

Skills employers value

Useful skills include:

- electronics fundamentals;
- electrical safety;
- basic mechanics;
- measurement and test-equipment discipline;
- schematic and technical-manual reading;
- troubleshooting logic;
- computer and networking fundamentals;
- documentation and work-order accuracy;
- preventive-maintenance discipline;
- calibration awareness;
- quality and change-control habits;
- infection-prevention awareness;
- cybersecurity awareness;
- communication with clinical and technical teams;
- customer service; and
- knowing when to stop and escalate.

The ability to explain a technical problem calmly to a nurse, therapist, physician, manager, vendor, or IT engineer can be as important as bench skills.

Education and entry pathway — United States

The U.S. Bureau of Labor Statistics reports that medical equipment repairers typically need an **associate's degree**. Biomedical technology, electronics, engineering technology, or a related field can provide relevant preparation. BLS also notes that some workers enter with a high-school diploma plus relevant training and work experience.

A practical U.S. pathway is:

1. Compare community-college biomedical/electronics programs in your region.
2. Review employer job postings to identify the equipment and skills most commonly requested locally.
3. Ask programs how much hands-on laboratory work is included.
4. Ask whether the program includes electronics, networking, troubleshooting, calibration, documentation, and medical-device safety.
5. Ask local hospitals, independent service organizations, manufacturers, and equipment vendors whether they hire trainees or provide structured on-the-job training.
6. Build safe hands-on experience with training equipment and approved labs rather than experimenting on active clinical devices.
7. Keep a portfolio of non-confidential troubleshooting exercises, measurements, diagrams, and maintenance documentation.

Do not pay for a private credential merely because an advertisement says it is "required everywhere." Verify actual employer requirements first.

United States — official income and outlook

The U.S. Bureau of Labor Statistics reports:

- May 2024 median annual wage: **USD $62,630**;
- median hourly wage: **USD $30.11**;
- 2024 employment: approximately **68,000**;
- projected 2034 employment: approximately **76,800**;
- projected growth, 2024–2034: **13%**;
- projected increase: approximately **8,800 jobs**; and
- approximately **7,300 openings per year** on average over the decade.

BLS also reports that in May 2024 the lowest 10 percent earned less than **USD $39,060**, while the highest 10 percent earned more than **USD $99,290**.

Selected industry median annual wages reported by BLS for May 2024 include:

- hospitals: **USD $74,560**;
- professional and commercial equipment/supplies merchant wholesalers: **USD $66,640**;
- electronic and precision equipment repair and maintenance: **USD $62,000**;
- ambulatory healthcare services: **USD $61,030**; and

- commercial/industrial machinery and equipment rental/leasing: **USD $42,650**.

These are national occupational statistics, not guaranteed starting pay. Local pay depends on geography, employer, experience, shift, travel, specialty, training, and other factors.

Free and lower-cost training support — United States

Before taking on private-school debt, investigate public and employer-supported options.

The CareerOneStop **American Job Center Finder** can help locate local workforce services:

Source link

CareerOneStop also provides a **WIOA-Eligible Training Program Finder**:

Source link

WIOA funding is not automatic. Eligibility depends on the individual, local workforce rules, program, provider, and available funding.

Also ask employers about paid trainee programs, tuition assistance, tool support, vendor training, certification reimbursement, or career-development benefits. Get repayment or service-commitment terms in writing before accepting employer-funded education.

Apprenticeship and work-based learning

Medical-equipment service pathways vary widely. Some employers develop technicians through structured trainee roles, field-service mentoring, internships, cooperative education, or apprenticeships; others expect prior electronics or biomedical education.

When evaluating work-based learning, ask:

- Is the training paid?
- Who supervises safety-critical work?
- What equipment may a trainee service?
- Are manufacturer courses included?
- Are tools and test equipment provided?
- How is competency documented?
- Are travel or on-call expectations included?
- Is there a repayment agreement?
- What role is available after training?

Do not confuse unpaid observation with a recognized apprenticeship.

Canada — related occupational family

The Government of Canada Job Bank lists **Biomedical and Laboratory Equipment Repairer** within a broader electrical/electronics engineering technologist and technician occupational family.

Job Bank reports that preparation in this broader family commonly involves college education, apprenticeship/training, or relevant experience depending on the exact role and jurisdiction. Competencies include troubleshooting, equipment/tool selection, quality-control testing, and preventive maintenance.

Because this is a broader occupational mapping, do not assume one Canadian title, credential, or provincial requirement applies to every biomedical-equipment role.

Canada — current wage evidence

Government of Canada Job Bank national wage data for the mapped occupational family report:

- low: **CAD $24.04/hour**;
- median: **CAD $35.58/hour**; and
- high: **CAD $55.34/hour**.

The wage table was updated November 19, 2025 using 2023–2024 reference data. The national occupation summary observed in 2026 also reports the **CAD $35.58/hour** national median.

Use provincial and local Job Bank filters before evaluating a specific offer.

Canada — credential and transfer warning

A U.S. degree, employer title, training certificate, or certification does not automatically create Canadian equivalence. Before relocating or paying for credential evaluation, verify:

- the exact NOC and job title used by the employer;
- provincial or territorial requirements;
- employer-specific education requirements;
- recognized college or apprenticeship pathways;
- language requirements; and
- immigration/work authorization separately from occupational qualification.

Colombia — SENA biomedical-equipment pathway

SENA official materials list **Mantenimiento de Equipo Biomédico** as a **Tecnólogo** program with a **24-month** duration.

A 2024 SENA circular lists **MANTENIMIENTO DE EQUIPO BIOMÉDICO**, Tecnólogo, presencial, at the Centro de Automatización Industrial in Caldas.

This is a strong public-training locator, but a normative or historical listing does not guarantee that a cohort is open today. Verify the current offering, center, schedule, admission requirements, modality, and availability through SENA/Betowa before making travel or spending decisions.

Official references:

Source link

Source link

Colombia — technovigilance and safety reporting

INVIMA's **Programa Nacional de Tecnovigilancia** is Colombia's post-market surveillance system for identifying, evaluating, managing, and preventing adverse events and incidents associated with medical devices.

Official source:

Source link

INVIMA specifically recognizes corrective-maintenance reports for biomedical equipment as a possible source for identifying problems where patient or operator health has been implicated.

A technician should therefore follow the institution's technovigilance, safety-event, quality, manufacturer, and regulatory reporting process when a malfunction may have harmed or endangered a patient or operator. Do not erase the evidence, silently repair the device, and return it to service without the required reporting and investigation.

Colombia — professional and employer boundaries

Completing a technology program does not authorize a worker to ignore:

- employer competency requirements;
- manufacturer service restrictions;
- electrical or occupational-safety rules;
- medical-device regulation;
- technovigilance requirements;
- metrology/calibration controls;
- radiation or other specialty authorization where applicable; or
- institutional quality and cybersecurity policies.

Verify the requirements of the exact job and equipment class before accepting responsibility.

Latin America and Caribbean exploration

Medical-device regulation, technical-education systems, occupational titles, and health-facility requirements differ by country.

A safe exploration sequence is:

1. Find the national medical-device regulator or health authority.
2. Identify public technical/vocational training institutions.
3. Search the regulator's device-maintenance, post-market surveillance, incident-reporting, and safety requirements.
4. Compare employer postings to identify common qualifications.
5. Verify whether the occupation or specific equipment class requires registration, authorization, radiation credentials, electrical licensing, metrology competency, or manufacturer training.
6. Confirm tuition and funding before paying a private provider.

Do not translate "BMET" or "medical equipment repairer" into a regulated title in another country without verification.

Choosing a training program

Before enrolling, ask:

- Is the institution recognized by the relevant education authority?
- What proportion of the program is hands-on?
- Which test equipment is used?
- Are electrical safety and calibration included?
- Are networking and cybersecurity included?
- Does the curriculum cover service documentation and quality systems?
- Does it explain the difference between servicing, modification, and remanufacturing?
- Are clinical-site placements available, and are they supervised?
- Are tools, uniforms, software, travel, exam fees, and other costs included?
- What employers have hired recent graduates?
- Can the school document those outcomes?
- Is there a refund policy?

Avoid schools that promise guaranteed employment, guaranteed immigration, universal certification, or unusually high starting salaries without verifiable evidence.

A practical 12-week exploration plan

Weeks 1–2 — understand the work

- Read the BLS occupation profile.
- Review ten local medical-equipment/BMET job postings.
- List common education, electronics, networking, travel, and on-call requirements.
- Identify which tasks are safety-critical.

Weeks 3–4 — build foundations

Study basic DC/AC electricity, Ohm's law, components, measurement, electrical safety, grounding, batteries, and safe use of a multimeter. Use training circuits, not active medical devices.

Weeks 5–6 — troubleshooting discipline

Practice fault isolation on low-risk training equipment. Build a repeatable method:

1. verify the complaint;
2. inspect safely;
3. check power and obvious faults;
4. consult approved documentation;
5. measure systematically;
6. record evidence;
7. repair only within scope; and
8. verify after repair.

Weeks 7–8 — digital and network fundamentals

Learn IP addressing, Ethernet, Wi-Fi basics, secure accounts, patching concepts, logs, backups, and change control. Focus on defensive support, not unauthorized testing.

Weeks 9–10 — healthcare quality and safety

Study preventive maintenance, calibration concepts, infection prevention, incident reporting, recall/alert handling, service records, and the FDA servicing/remanufacturing distinction.

Weeks 11–12 — career decision

- Compare two or three credible training options.
- Check public funding and employer sponsorship first.
- Speak with technicians or clinical-engineering departments when possible.
- Build a non-confidential portfolio.

- Apply to realistic trainee, electronics, field-service, or biomedical-equipment roles.

Building experience before the first biomedical role

Transferable experience can come from:

- electronics repair;
- industrial maintenance;
- instrumentation;
- calibration laboratories;
- field service;
- IT/network support;
- military electronics/medical-equipment specialties;
- laboratory equipment support;
- manufacturing test; or
- technical customer support.

Describe transferable skills accurately. Do not claim medical-device service experience if your work was only consumer electronics or general IT.

Portfolio ideas

A safe portfolio can include:

- low-voltage troubleshooting exercises;
- block diagrams;
- sample preventive-maintenance checklists for fictional training equipment;
- calibration-concept notes;
- anonymized work-order examples;
- network troubleshooting labs;
- cybersecurity change-control examples;
- risk-assessment exercises; and
- short explanations of when to escalate rather than repair.

Never put real patient data, employer-confidential material, proprietary manuals, device keys, or unauthorized service information in a public portfolio.

Résumé language that stays accurate

Examples of supportable language when true:

- "Performed systematic troubleshooting using approved technical documentation and test equipment."

- "Documented preventive and corrective maintenance with traceable measurements and parts records."
- "Applied electrical-safety, change-control, and escalation procedures for safety-sensitive equipment."
- "Supported network-connected equipment while following credential and cybersecurity controls."
- "Coordinated technical issues with users, vendors, engineering, IT, and quality teams."

Do not use "certified," "licensed," "manufacturer-authorized," or "clinical engineer" unless the statement is factually correct for you and the jurisdiction.

Interview questions for an employer

Ask:

- What equipment will I support?
- What tasks require manufacturer training?
- What is the escalation path for modifications or unusual repairs?
- How are preventive-maintenance intervals determined?
- What test equipment and calibration controls are used?
- What cybersecurity responsibilities belong to this role?
- How are recalls, safety alerts, and technovigilance/adverse events handled?
- How much travel or on-call work is expected?
- What training is provided during the first six months?
- Are tools, travel, phone, and certification/training costs reimbursed?

Accessibility and disability support

People with disabilities can work successfully in technical service when job tasks and accommodations align.

Possible supports may include:

- ergonomic benches and adjustable work surfaces;
- magnification or improved lighting;
- screen readers or speech-to-text for documentation;
- hearing-access supports;
- written procedures and checklists;
- task sequencing aids;
- accessible test equipment or software interfaces;
- schedule adjustments where legally and operationally feasible; and
- alternative methods for non-essential physical tasks.

An accommodation does not require bypassing an essential safety control. Discuss essential functions and accommodation options through the appropriate employer or education process.

Scam detection and spending-pause checks

Pause before paying if a school, recruiter, or credential vendor:

- promises a guaranteed job;
- promises a guaranteed U.S., Canadian, or Colombian work visa;
- claims one certificate is legally required everywhere;
- pressures you to pay immediately;
- refuses to identify the accrediting/recognizing authority;
- cannot show the curriculum or total cost;
- advertises implausible starting wages without local evidence;
- asks for payment by gift card, cryptocurrency, or unusual transfer; or
- tells you not to verify claims with employers or public agencies.

Verify first. Public institutions and employers are often better sources of requirement information than marketing pages.

Career progression

Possible progression depends on education, performance, employer structure, and jurisdiction. Paths may include:

- senior biomedical equipment technician;
- imaging or laboratory-equipment specialist;
- field service engineer/technician;
- clinical engineering support;
- technical trainer;
- service coordinator;
- quality or regulatory support;
- cybersecurity-focused medical-device support;
- supervisor or manager; or
- further engineering/technology education.

A title such as "engineer" may be regulated in some jurisdictions. Do not claim a protected professional title without verifying local rules.

Source list — verify before important decisions

United States

U.S. Bureau of Labor Statistics — Medical Equipment Repairers
Source link

U.S. Food and Drug Administration — Remanufacturing and Servicing Medical Devices
Source link

U.S. Food and Drug Administration — Remanufacturing of Medical Devices, final guidance
Source link

FDA — Cybersecurity
Source link

FDA — Postmarket Management of Cybersecurity in Medical Devices
Source link

CareerOneStop — American Job Center Finder
Source link

CareerOneStop — WIOA-Eligible Training Program Finder
Source link

Canada

Government of Canada Job Bank — Biomedical And Laboratory Equipment Repairer
Source link

Government of Canada Job Bank — wages
Source link

Colombia

SENA — Resolución 472 de 2014, program listing
Source link

SENA — Circular 22 de 2024, training-center/program listing
Source link

INVIMA — Programa Nacional de Tecnovigilancia
Source link

Final reality check

Medical-equipment repair can be a strong technical healthcare career for people who like troubleshooting, precision, documentation, electronics, and service. It is also safety-critical work. A rushed repair, undocumented change, unauthorized firmware update, missed calibration problem, concealed incident, or cybersecurity mistake can have consequences far beyond an ordinary electronics failure.

The strongest technicians are not the people who always say "I can fix it." They are the people who know how to verify, document, test, protect the patient, and stop when the work crosses their authorized boundary.